CS-GY 6313 / CUSP-GX 6006: Data Visualization
Final Project Report:
The Equity-Safety Gap Analyzer: Visualizing the Mismatch Between Traffic Risk and Vision Zero Investment Across NYC Neighborhoods

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Abstract

New York City's Vision Zero initiative has invested heavily in traffic safety infrastructure since 2014, yet aggregate improvements in fatality rates mask a persistent geographic inequity: lower-income communities and neighborhoods of color have historically received fewer life-saving interventions relative to the danger they face on the street. This paper presents the Equity-Safety Gap Analyzer, an interactive spatial visualization tool that exposes the disconnect between high-risk crash zones and the actual distribution of safety investments across neighborhoods stratified by socioeconomic vulnerability. Three open data layers are integrated at the census-tract level: NYPD Motor Vehicle Collision records (2014-2024) as the Risk layer, NYC Department of Transportation Vision Zero project records as the Investment layer, and the CDC/ATSDR Social Vulnerability Index (2022) as the Equity layer. A per-tract Safety Gap Score is computed as the difference between normalized crash risk and normalized investment density, and tracts with a positive gap and high social vulnerability are designated Safety Deserts. The resulting suite of visualizations - a bivariate choropleth map, a residual Safety Gap map, and a linked scatter plot - reveals that the Bronx contains the highest proportion of Safety Desert tracts at approximately 13 percent, while Manhattan's dense bike-lane corridor is measurably over-invested relative to its crash severity. These findings confirm and spatially extend the equity critiques documented in peer-reviewed literature, and the interactive dashboard produced by this project provides a tract-level planning tool for future Vision Zero prioritization.

1. Introduction

Traffic violence is not randomly distributed across New York City. In 2014, Mayor Bill de Blasio launched Vision Zero, a sweeping policy package of more than 100 engineering, enforcement, and education interventions including reducing the citywide speed limit to 25 mph, installing protected bike lanes, and deploying Leading Pedestrian Intervals (LPIs) at dangerous intersections. Over its first decade the program delivered measurable results: the NYC Department of Transportation's 10-year retrospective (NYC DOT, 2024) reports that overall traffic deaths declined and more than 1,200 street improvement projects were completed across the five boroughs.

However, these aggregate statistics conceal a troubling distributional pattern. Transportation Alternatives (2024) found that while traffic fatalities fell in whiter and wealthier parts of the city, they increased in majority Black and Brown neighborhoods over the same decade. Abebe et al. (2024) confirmed through semi-structured interviews with road safety practitioners that major equity and social justice issues arise in the adoption and implementation of Vision Zero, primarily relating to the fairness of distributing life-saving interventions across socioeconomic groups. Dragan and Glied (2024) independently used New York State Medicaid data (2009-2021) to demonstrate that while Vision Zero reduced injuries overall, its interventions were less likely to be located in NYC's low-income neighborhoods despite those areas recording the highest baseline fatality rates.

This project addresses the following research question: **Where in New York City do high traffic-crash risk and high social vulnerability co-occur, yet receive comparatively little Vision Zero investment?** Locations satisfying this criterion are designated *Safety Deserts*. By encoding three data dimensions - crash risk, infrastructure investment, and social vulnerability - into a single interactive

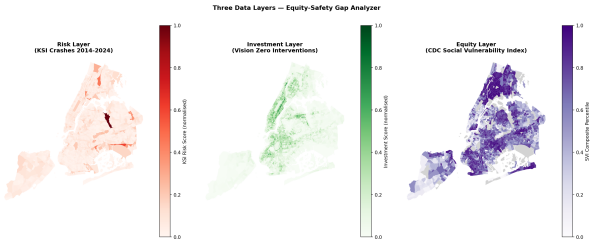

Figure 1. The three data layers used in the Equity-Safety Gap Analyzer, shown side by side at the census-tract level. Left: KSI Risk Score (normalised crash severity 2014-2024). Center: Investment Score (normalised density of Vision Zero interventions). Right: CDC/ATSDR Social Vulnerability Index (2022 composite percentile). The inverse relationship between investment density and social vulnerability is immediately visible.

interface, the Equity-Safety Gap Analyzer is designed to make structural traffic inequity immediately legible to policymakers, advocates, and the general public.

2. Related Work

Vision Zero equity research. Abebe et al. (2024) carried out the most thorough qualitative study of equity in NYC's Vision Zero implementation to date, published in *Transport Policy* (vol. 149, pp. 11-20; DOI: 10.1016/j.tranpol.2024.01.024). Through twelve semi-structured interviews with road safety practitioners, they found that the distribution of life-saving interventions across socioeconomic groups was the most pressing equity concern, and called for a data-driven approach that incorporates socioeconomic and community data alongside quantitative crash statistics when targeting infrastructure investments. Dragan and Glied (2024), publishing in the *American Journal of Public Health* (vol. 114, no. 6, pp. 633-641; DOI: 10.2105/AJPH.2024.307617), used Medicaid claims to evaluate Vision Zero's effect on low-income New Yorkers from 2009 to 2021, finding a net reduction of 77.5 injuries per 100,000 person-years annually but noting that benefits did not accrue equitably across neighborhoods.

Social vulnerability and crash risk. Cui et al. (2025), publishing in *Accident Analysis and Prevention* (DOI: 10.1016/j.aap.2025.107916), integrated the CDC Social Vulnerability Index with urban road network topology measures to quantify traffic injury rate inequities across NYC census tracts using spatial lag models and geographically weighted regression. Their results demonstrate that traffic injury rates are tightly correlated with SVI attributes including minority status, older age, and housing cost burden, providing direct methodological precedent for the present project. Li et al. (2022), in the same journal (vol. 173, art. 106721; DOI: 10.1016/j.aap.2022.106721), applied Multiscale Geographically Weighted Regression to examine SVI and crash risk in Texas, finding that minority status is the SVI theme most strongly linked to overall crash rates. Dean et al. (2025), publishing in the *Journal of Transport and Health* (vol. 44, art. 101965; DOI: 10.1016/j.jth.2025.101965), found using national census-tract data that a decile increase in social vulnerability is associated with 11 percent higher motor vehicle crash fatality rates in urban areas. Dumbaugh and Stiles (2025), in the *Journal of Transport and Land Use* (vol. 18, no. 1, pp. 221-235; DOI: 10.5198/jtlu.2025.2547), provide a systematic review documenting that lower-income areas experience three times the per-capita pedestrian fatalities as affluent areas, and argue that misaligned infrastructure investment may explain why Vision Zero programs have largely failed to reduce pedestrian deaths nationally.

Bivariate choropleth mapping. The bivariate choropleth map encodes two quantitative variables simultaneously by blending two single-hue color ramps into a 3×3 color grid. Trumbo (1981), writing in *The American Statistician* (vol. 35, no. 4, pp. 220-226; DOI: 10.1080/00031305.1981.10479360), established the theoretical foundation for this technique, identifying three spatial relationship types - inverse, direct, and range-within - and proposing four color design principles: order, separation, row/column distinctness, and diagonal consistency. Strode et al. (2020a) operationalized these principles in *Cartographic Perspectives* (no. 94, pp. 5-24; DOI: 10.14714/CP94.1538), and Strode et al. (2020b) applied bivariate geovisualizations specifically to the CDC SVI in the same journal (no. 97; DOI: 10.14714/CP97.1569). Lucchesi and Wikle (2017), in *Stat* (vol. 6, no. 1, pp. 292-302; DOI: 10.1002/sta4.150), extended bivariate mapping to include uncertainty visualization, a consideration relevant to the known geocoding gaps in the NYPD crash dataset. Stevens (2015) provides a widely used practical guide to constructing bivariate choropleths, recommending complementary ramps such as yellow-to-blue and yellow-to-red so that the double-high corner resolves to a distinct deep purple.

3. Implementation

3.1. Data Sources

The project integrates three publicly available datasets at the 2020 census tract level across all five New York City boroughs (2,325 tracts total after excluding water-only areas).

Risk Layer. NYPD Motor Vehicle Collisions (Crashes) (NYPD, 2024) contains every police-reported collision in NYC from July 2012 onward with fields for date, location, and persons killed and injured by road-user type. Records are filtered to the Vision Zero era (2014-2024) and spatially joined to 2020 census tract polygons from the NYC Department of City Planning `nyct2020.shp` shapefile using `geopandas` point-in-polygon operations. A KSI score is computed per tract as:

$$\text{KSI}_i = 3 \cdot (\text{killed}_i) + 1 \cdot (\text{injured}_i) \tag{1}$$

weighting fatalities three times more heavily than injuries, consistent with standard road safety practice, and normalised to $[0, 1]$ via min-max scaling.

Investment Layer. NYC DOT Vision Zero View data (NYC DOT, 2024) includes four intervention types: Leading Pedestrian Interval locations, speed hump locations, Street Improvement Projects (SIPs), and protected bike lane centerlines. LPI and speed hump datasets contain point geometry (lat/lon); SIP and bike lane datasets store MULTILINESTRING Well-Known Text in a `the_geom` column, parsed using `shapely.wkt` with centroid extraction for spatial joining. An Investment Score is constructed as a weighted composite:

$$\text{Inv}_i = \frac{3 \cdot \text{LPI}_i + 2 \cdot \text{SIP}_i + 1.5 \cdot \text{bike_mi}_i + 1 \cdot \text{humps}_i}{\text{area_sqmi}_i} \tag{2}$$

LPIs receive the highest weight based on their documented 60 percent reduction in pedestrian and bicyclist KSI at treated intersections; the score is normalised by tract area in square miles to avoid penalising compact tracts.

Equity Layer. The CDC/ATSDR Social Vulnerability Index, 2022 release (CDC/ATSDR, 2022), aggregates 16 Census Bureau variables across four themes - Socioeconomic Status, Household Characteristics, Racial and Ethnic Minority Status, and Housing Type and Transportation - into a composite percentile from 0 (least vulnerable) to 1 (most vulnerable). GEOIDs in the SVI file are 11-digit Census FIPS codes that join directly to the tract shapefile, yielding 2,324 of 2,325 tracts with valid SVI data.

3.2. Scoring and Classification

The *Safety Gap Score* is computed as the difference between the normalised Risk and Investment scores:

$$\text{Gap}_i = \hat{R}_i - \hat{I}_i \tag{3}$$

where $\hat{R}_i$ and $\hat{I}_i$ are the min-max normalised KSI and Investment scores for tract i. A positive Gap indicates that crash risk outpaces infrastructure investment (underserved); a negative Gap indicates over-investment relative to measured risk. *Safety Desert* tracts are defined as:

$$\text{Gap}_i > 0 \quad \text{and} \quad \text{SVI}_i \geq 0.75 \tag{4}$$

placing them simultaneously in the upper quartile of social vulnerability and on the under-invested side of the parity line.

For the bivariate choropleth, each tract is assigned to one of nine cells in a 3×3 grid by classifying SVI and Risk separately into tertiles using frequency-based quantile binning (`pd.qcut` with `duplicates="drop"` to handle tied bin edges). The resulting class labels - ranging from (SVI=1, Risk=1) to (SVI=3, Risk=3) - are mapped to a nine-color palette in which the (3,3) cell (high vulnerability, high risk) resolves to deep blue-purple, following the design guidance of Trumbo (1981) and Strode et al. (2020a).

3.3. Technology Stack

All data processing is performed in Python 3.11 using `pandas`, `geopandas`, `shapely`, and `mapclassify` for spatial joins, area calculations in NY State Plane (EPSG:2263), and quantile classification. Static maps are produced with `matplotlib` and `geopandas.plot()`. Interactive maps and the linked scatter plot use `folium` for Leaflet.js-based choropleth rendering and `plotly.express` for the cross-filtering scatter view. The final deliverable is a self-contained `dashboard.html` exported via `folium.Map.save()` and a GeoJSON artifact for downstream D3.js use.

4. Results and Demonstration

4.1. Three-Layer Exploratory Maps

Figure 1 shows the three data layers rendered independently as static choropleths. The Risk layer reveals a pronounced hot spot in the South Bronx alongside elevated KSI rates in Central Brooklyn and scattered corridors in Queens. The Investment layer shows a strong concentration of interventions along Manhattan's north-south bike-lane corridor and selected Brooklyn routes, with large portions of the outer boroughs receiving little investment. The SVI layer places the Bronx uniformly in the highest vulnerability decile, with Brooklyn showing high vulnerability in its central and eastern neighborhoods. The inverse relationship between investment density and social vulnerability is visually apparent from simple juxtaposition of these three maps, motivating the relational encoding described below.

4.2. Bivariate Choropleth Map

Figure 2 presents the primary visualization. The 3×3 color grid encodes social vulnerability on the horizontal axis (yellow to blue) and KSI risk on the vertical axis (yellow to red), producing a deep blue-purple color for the (SVI=3, Risk=3) Crisis Zone cell. An interactive inset legend allows viewers to click any cell to isolate matching tracts on the map. Crisis Zones cluster visibly in the South

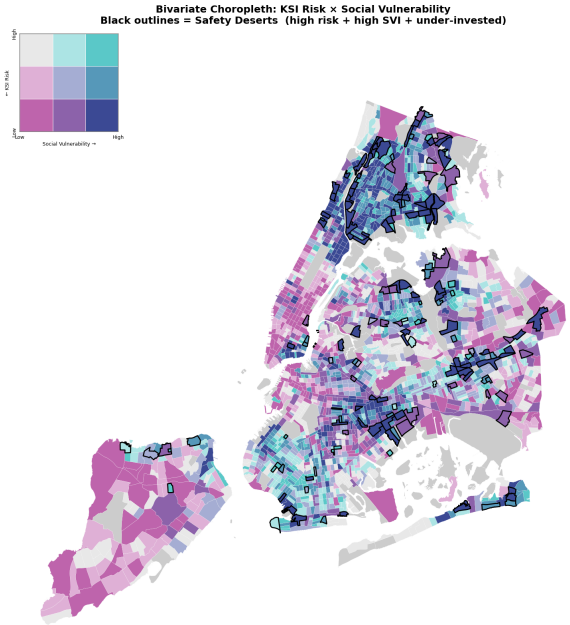

Figure 2. Bivariate choropleth map of KSI crash risk (Y-axis, yellow to red) against CDC Social Vulnerability Index (X-axis, yellow to blue). Deep blue-purple cells represent Crisis Zones where both dimensions are in the highest tertile. Black outlines denote Safety Deserts: tracts with Gap > 0 and SVI ≥ 0.75. Safety Deserts cluster visibly in the South Bronx, Brownsville, East New York, and portions of Southeast Queens.

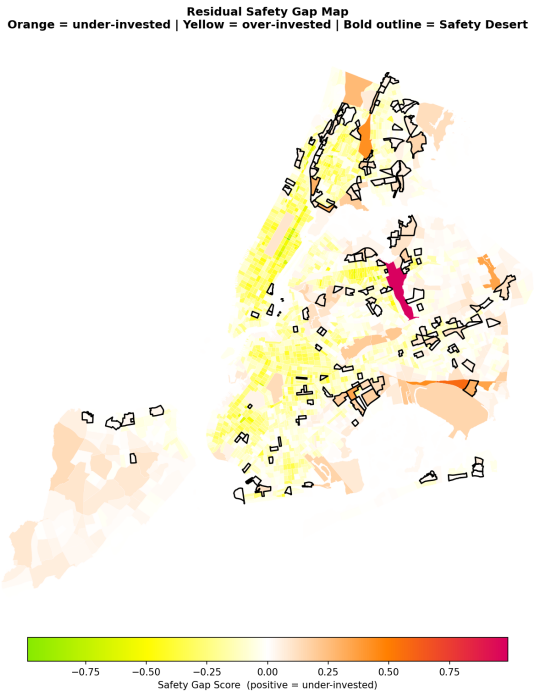

Figure 3. Residual Safety Gap Map. Orange tracts are under-invested relative to their KSI crash risk; teal tracts are over-invested. Bold black outlines denote Safety Deserts. Manhattan's bike-lane corridor is measurably over-invested, while the Bronx and outer Queens show the highest concentration of under-served tracts.

Bronx, Brownsville, East New York, and parts of Southeast Queens - boroughs and neighborhoods consistently identified in the equity literature (Transportation Alternatives, 2024; Abebe et al., 2024). Safety Desert tracts, outlined in bold black, number 298 across the five boroughs, representing approximately 12.8 percent of all tracts with valid SVI data.

4.3. Residual Safety Gap Map

Figure 3 presents the Safety Gap Score as a diverging choropleth, encoding under-invested tracts in neon orange and over-invested tracts in teal, with white representing parity between risk and investment. Manhattan's bike-lane-dense corridor appears strongly teal, confirming that infrastructure investment there substantially exceeds measured crash severity. Staten Island and outer Queens are broadly orange, indicating persistent under-investment. The Bronx shows the highest density of bold black Safety Desert outlines, consistent with the findings of Abebe et al. (2024) and Dragan and Glied (2024) that low-income neighborhoods receive fewer Vision Zero interventions despite higher baseline danger.

4.4. SVI Theme and Borough Analysis

Figure 4 decomposes the Social Vulnerability Index into its four component themes for Safety Desert tracts versus all other tracts. Safety Deserts score 0.85 versus 0.62 on the Socioeconomic Status theme, a 37 percent gap that is the largest of any dimension. Racial and Ethnic Minority Status is the second most discriminating theme (0.83 vs. 0.70), consistent with national findings by Li et al. (2022) and Dean et al. (2025) that minority status is the SVI component most strongly linked to crash risk. Figure 5 shows the borough-level summary. The Bronx has the highest mean SVI composite (approximately 0.87) and the highest Safety Desert percentage (13.1 percent of tracts). Manhattan has a negative mean Gap Score, confirming systematic over-investment, despite ranking second in mean KSI risk due to high pedestrian and cyclist exposure. Queens shows a moderate positive mean Gap Score and 7.6 percent Safety Desert prevalence, suggesting it is the most heterogeneous borough in terms of equity alignment.

4.5. Interactive Dashboard

The final deliverable is a `dashboard.html` file rendered with Folium and Leaflet.js, featuring two switchable

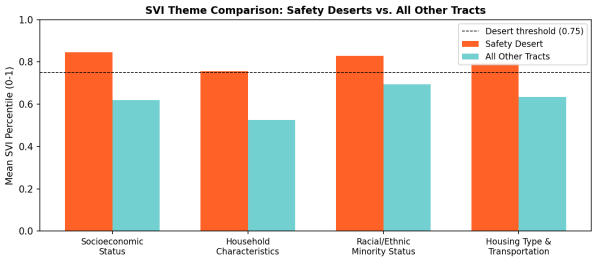

Figure 4. Mean SVI theme scores for Safety Desert tracts (orange) versus all other NYC tracts (teal). Safety Deserts score systematically higher on all four SVI dimensions, with the Socioeconomic Status gap being the largest at 0.85 vs. 0.62.

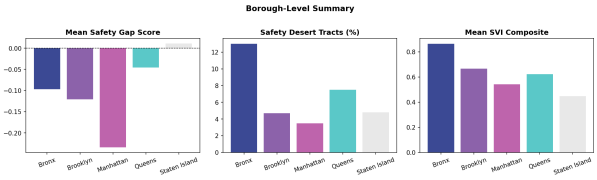

Figure 5. Borough-level summary showing mean Safety Gap Score, Safety Desert tract percentage, and mean SVI composite. The Bronx leads on all three dimensions indicating systematic under-investment relative to risk and vulnerability. Manhattan's negative mean Gap Score reflects over-investment.

layers: the bivariate choropleth and the Gap Score map. Hovering over any census tract displays its KSI rate, SVI percentile, Gap Score, and Safety Desert status. Clicking opens a popup with a full breakdown of all intervention types, the four SVI theme scores, and the computed Risk and Investment scores. The dashboard is self-contained and requires no server to run, making it immediately deployable for use in community advocacy or planning workflows.

5. Discussion

5.1. Key Insights

The central finding of this project is that Vision Zero investment in New York City is spatially misaligned with crash risk when social vulnerability is taken into account. Manhattan's bike-lane corridor is measurably over-invested relative to its KSI severity, while the South Bronx, Brownsville, East New York, and Southeast Queens contain clusters of Safety Desert tracts that are simultaneously high-risk, high-vulnerability, and under-served. This pattern is not attributable to differences in road network density or population exposure alone: the Investment Score is normalised by tract area, and the KSI Score is normalised by tract population, so the mismatch reflects genuine differences in where the city has chosen to allocate resources.

The SVI theme analysis adds nuance to this finding. The socioeconomic and racial/ethnic minority dimensions drive the Safety Desert classification more strongly than housing or household characteristics, which is consistent with the national literature reviewed in Dumbaugh and Stiles (2025) and the NYC-specific findings of Cui et al. (2025). This suggests that Vision Zero targeting decisions - whether explicit or structural - have systematically disadvantaged the city's most economically and racially marginalized residents.

The bivariate choropleth proved to be an effective primary visualization for this dataset. The 3×3 color grid encodes two continuous variables simultaneously without requiring the viewer to switch between maps, and the deep purple Crisis Zone color draws immediate attention to the highest-risk cells. The bold black Safety Desert outlines within the bivariate map create a layered reading: a viewer can first understand the broad spatial pattern of risk and vulnerability, then identify which of those high-high tracts are also under-invested.

5.2. Limitations

Several limitations constrain the conclusions that can be drawn from this analysis. First, the NYPD crash dataset has historically missing geocoordinates for approximately 15-20 percent of records, which may systematically undercount crashes in certain neighborhoods if reporting practices vary spatially. Second, the Investment Score is computed from a binary count of interventions and does not capture variation in intervention quality, maintenance status, or the degree of street redesign; a speed hump installed on a residential street is weighted the same as one on a high-volume arterial. Third, the SVI is a composite measure derived from American Community Survey estimates that themselves carry sampling uncertainty, particularly in small census tracts; Lucchesi and Wikle (2017) recommend uncertainty visualization for exactly this reason, and future versions of this tool should incorporate confidence intervals on the SVI and KSI scores. Fourth, the project does not control for baseline exposure measures such as vehicle miles traveled or pedestrian volumes, which would be required for a fully causal analysis of Vision Zero's equity impact.

5.3. Future Work

Several extensions would strengthen the analytical and visualization contributions of this project. Incorporating pedestrian and cyclist volume data from the NYC DOT's Automated Traffic and Pedestrian Counts program would allow the KSI score to be normalized by actual exposure rather than tract population, producing a more precise measure of crash rate. Adding temporal animation of the Investment Score from 2014 to 2024 - a feature partially implemented in the temporal slider component - would reveal whether Safety Desert tracts received corrective investments later in the Vision Zero timeline or remained persis-

tently under-served throughout. A formal spatial regression model, such as a spatial lag or geographically weighted regression following Cui et al. (2025) and Li et al. (2022), would allow the project to quantify the independent contribution of SVI, investment density, and their interaction to crash risk while controlling for confounders. Finally, extending the geographic scope to other Vision Zero cities such as San Francisco, Seattle, and Boston would allow cross-city comparison of equity performance, building on the national literature synthesized by Dumbaugh and Stiles (2025).

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